COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212- -	**SECTION:**	CPE31S2
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ACTIVITY NO. 2.1 SSH Key-Based Authentication and GIT Setup

1. PROCEDURE OUTPUT

Task 1: Create an SSH Key Pair for User Authentication

```
duenas@workstation:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/duenas/.ssh/id_ed25519): id_rsa
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in id_rsa
Your public key has been saved in id_rsa.pub
The key fingerprint is:
SHA256:sAw0ek2VeDhrwDSIDo951oyeArHOmNA6fmT9SQZKWg8 duenas@workstation
The key's randomart image is:
+--[ED25519 256]--+
| . += .+..       |
|+ .oo=+ o        |
|oB.+o.o+         |
|*.*Eo+oo         |
|**+.=.+ S        |
|*+o+ o o         |
|.oo   + .        |
| . .   o         |
|  .              |
+----[SHA256]-----+
duenas@workstation:~$
```

```
duenas@workstation:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.

Enter file in which to save the key (/home/duenas/.ssh/id_rsa): Enter passphrase
 (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/duenas/.ssh/id_rsa
Your public key has been saved in /home/duenas/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:ugy8FKFt1Ryc2sLBr5DPAP59i2t4PC7eADxmyOswMWc duenas@workstation
The key's randomart image is:
+---[RSA 4096]----+
|     . ...       |
|  .   ooo.       |
| . ..o.=o        |
|.o.o+o+ o        |
|+.E.+* oS        |
| B.=..=..        |
|o.  =o.o .       |
|o. .o**..        |
| . .o=*o         |
+----[SHA256]-----+
duenas@workstation:~$
```

```
duenas@workstation:~$ ls -la .ssh
total 24
drwx------  2 duenas duenas 4096 Jul 30 00:22 .
drwxr-x--- 15 duenas duenas 4096 Jul 30 00:20 ..
-rw-------  1 duenas duenas    0 Jul  9 02:14 authorized_keys
-rw-------  1 duenas duenas 3434 Jul 30 00:22 id_rsa
-rw-r--r--  1 duenas duenas  744 Jul 30 00:22 id_rsa.pub
-rw-------  1 duenas duenas 1404 Jul 16 02:13 known_hosts
-rw-r--r--  1 duenas duenas  142 Jul 16 01:56 known_hosts.old
```

Task 2: Copying the Public Key to the remote servers

```
duenas@workstation:~$ ssh-copy-id
Usage: /usr/bin/ssh-copy-id [-h|-?|-f|-n|-s|-x] [-i [identity_file]] [-p port] [
-F alternative ssh_config file] [-t target_path] [[-o <ssh -o options>] ...] [us
er@]hostname
        -f: force mode -- copy keys without trying to check if they are already
installed
        -n: dry run    -- no keys are actually copied
        -s: use sftp   -- use sftp instead of executing remote-commands. Can be
useful if the remote only allows sftp
        -x: debug      -- enables -x in this shell, for debugging
        -h|-?: print this help
```

```
duenas@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa duenas@workstation
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/duenas/.ssh
/id_rsa.pub"
The authenticity of host 'workstation (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:mTRnDWLxPAlo/UyJkKj6Rd26mWUOSw3XchH9h5iGg3E.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
    ~/.ssh/known_hosts:5: [hashed name]
    ~/.ssh/known_hosts:6: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
duenas@workstation's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'duenas@workstation'"
and check to make sure that only the key(s) you wanted were added.

duenas@workstation:~$
```

```
duenas@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa duenas@server1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/duenas/.ssh
/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
duenas@server1's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'duenas@server1'"
and check to make sure that only the key(s) you wanted were added.
```

```
duenas@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa duenas@server2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/duenas/.ssh
/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
duenas@server2's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'duenas@server2'"
and check to make sure that only the key(s) you wanted were added.
```

Reflections:
Answer the following:

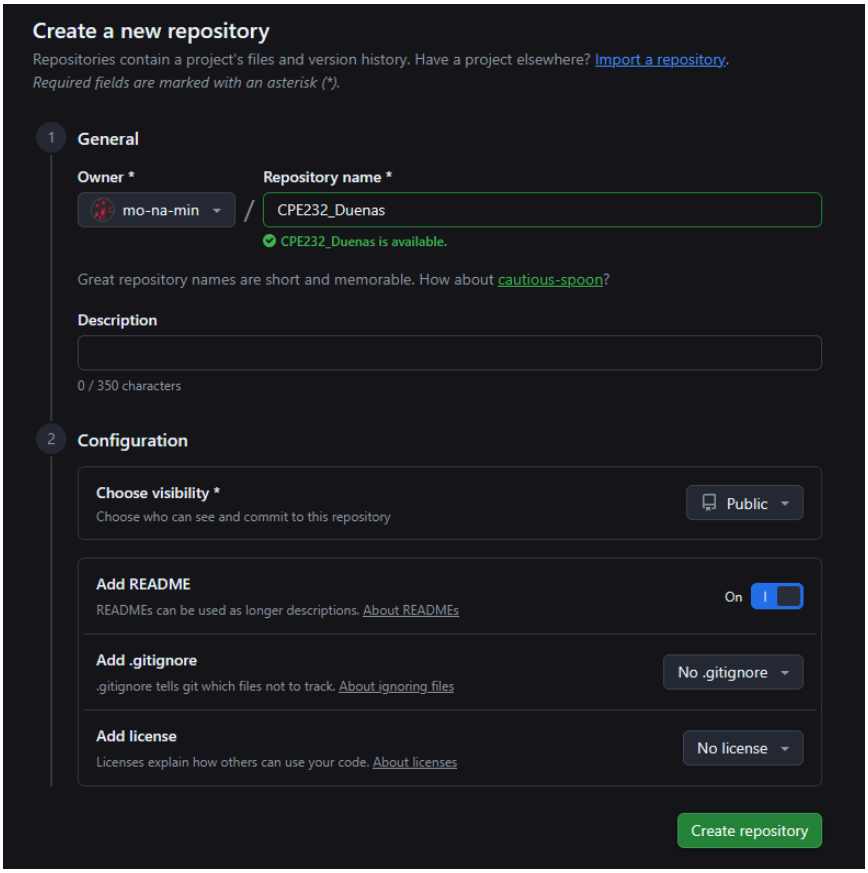
1. How will you describe the ssh-program? What does it do?
   - It is a tool in which it encrypts the datas and files that you choose, making it secured from malicious attacks.
2. How do you know that you already installed the public key to the remote servers?
   - Run ssh -i /path/to/private_key username@host. If it doesn't ask for password then it is already installed on your end.

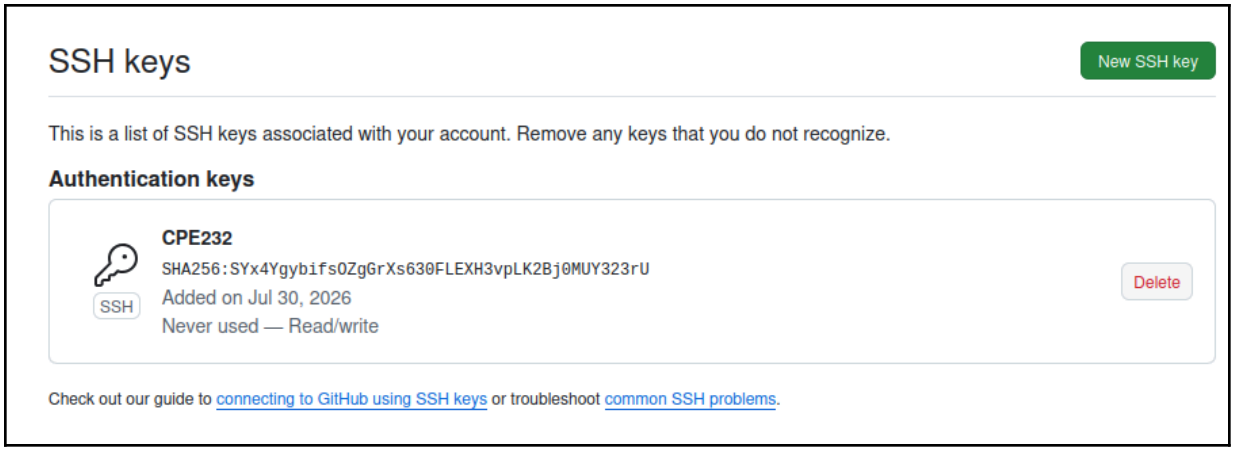
Part 3: Discussion
Set up Git

```
duenas@workstation:~$ which git
duenas@workstation:~$ sudo apt install git
[sudo] password for duenas:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
duenas@workstation:~$ which git
/usr/bin/git
```

```
duenas@workstation:~$ git --version
git version 2.43.0
```








```
duenas@workstation:~/CPE232_Duenas$ git config --global user.email "ranzelaldous
gabrielduenas@gmail.com"
duenas@workstation:~/CPE232_Duenas$ git config --global user.name "mo-na-min"
```

```
duenas@workstation:~$ git clone git@github.com:mo-na-min/CPE232_Duenas.git
Cloning into 'CPE232_Duenas'...
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
duenas@workstation:~$ ls
CPE232         Desktop    Downloads  id_rsa.pub  Pictures  snap       Videos
CPE232_Duenas  Documents  id_rsa     Music       Public    Templates
```

```
duenas@workstation:~/CPE232_Duenas$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

```
duenas@workstation:~/CPE232_Duenas$ git commit -m "hello world"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

```
duenas@workstation:~/CPE232_Duenas$ git push origin main
Everything up-to-date
```

Reflections:
Answer the following:
3. What sort of things have we so far done to the remote servers using ansible commands?
   - We haven't executed any Ansible commands in this session yet, as we've only set up SSH keys and Git account configurations.
4. How important is the inventory file?
   - The inventory file is essential because it defines the exact list of remote hosts and IP addresses that Ansible is meant to manage.

2. Conclusion

We resolved an issue where GitHub rejected an invalid SSH key by generating an Ed25519 key and retrieving its exact contents directly through the terminal. Following that, we configured Git's global user identity and pushed your local commits to your remote GitHub repository using git push origin main.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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